A CASE-STUDY BASED PRESENTATION | MACHINE LEARNING | B.TECH | A.Y. 2026-27

Crop Recommendation Using Multinomial Logistic Regression

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Introduction

Background

- Crop choice decides yield, income and food supply
- Each crop needs its own nutrients, temperature, humidity, rainfall and pH
- A wrong choice lowers productivity and causes financial loss

Domain overview

- Machine learning in agriculture (smart farming)
- Supervised multi-class classification
- Multinomial Logistic Regression predicts 1 of 22 crops

Why it is needed

- Farmers often decide from experience or neighbours' practice
- A quick, data-driven recommendation is needed
- Class probabilities make the advice easy to explain

Problem Statement

Farmers lack a quick, accurate, data-driven way to decide which of many crops suits their soil and climate.

Existing problem

- Crop selection relies on experience and local practice
- Soil nutrients and weather are rarely weighed together
- Mistakes show up only after the season

Limitations / challenges

- Human judgement is subjective and inconsistent
- Seven interacting factors are hard to weigh by hand
- 22 candidate crops make the choice complex

Who is affected

- Farmers, especially small and first-time cultivators
- Agricultural advisors and extension workers
- Consumers, through lower yields and supply

Motivation



Why we selected this project

Agriculture is a high-impact use of machine learning, and crop recommendation is a clean multi-class problem we could study end to end: data, EDA, model and evaluation.



Real-world need

Farmers need a fast decision-support tool that uses soil nutrients (N, P, K, pH) and climate (temperature, humidity, rainfall) together instead of guesswork.



Research / industry motivation

Precision and smart farming increasingly rely on data-driven recommendations. Logistic regression is a fast, interpretable baseline that also gives class probabilities.

Objectives

MAIN OBJECTIVE Build a Multinomial Logistic Regression model that recommends the most suitable crop from seven soil and climate parameters with high accuracy.

1. Load, inspect and clean the crop dataset
2. Explore the data with EDA to see crop-wise conditions
3. Standardise features and make a stratified 80/20 split
4. Train the model and tune C by 5-fold cross-validation
5. Evaluate with accuracy, precision, recall, F1 and confusion matrix

Existing System



Farmer experience & local practice





Rough weather / soil-test check





Crop decision

Current approach / technologies

- Decisions from farmer experience
- Local practice and neighbours' choices
- Crop calendars and rules of thumb
- Manual soil testing and advisory visits

Limitations

- Subjective and inconsistent
- Does not combine all seven factors
- No probability or confidence in the advice
- Slow and not easily available to every farmer

Proposed System



Our solution

- Multinomial Logistic Regression crop recommender
- Inputs: N, P, K, temperature, humidity, pH, rainfall
- Features standardised, model tuned by cross-validation
- Output: recommended crop and probability for each crop

Subjective decisions	Learns from 2,200 labelled records
Factors weighed separately	Uses all 7 factors together in one model
No confidence information	Returns a probability for each crop
Slow, manual advice	Trains in under a second; instant prediction

System Architecture

INPUT

- Soil: N, P, K, pH
- Climate: temperature, humidity, rainfall
- Crop dataset (CSV, 2,200 records)

PROCESSING

- Data inspection and cleaning
- Standardisation (StandardScaler)
- Stratified 80/20 train-test split

MODEL

- Multinomial Logistic Regression (softmax, L2)
- C tuned by 5-fold CV
- Best C = 100

OUTPUT

- Recommended crop (1 of 22)
- Probability for each crop
- Accuracy, precision, recall, F1, confusion matrix

Methodology / Workflow

1
Collect data
Load the crop CSV with Pandas

2
Inspect & clean
Shape, types, nulls, duplicates

3
Explore (EDA)
Distributions, heatmap, box plots

4
Standardise
Zero mean, unit variance

5
Split data
Stratified 80/20 split

6
Train model
Multinomial Logistic Regression

7
Tune C
5-fold cross-validation

8
Evaluate
Metrics, matrix, new prediction

Technology Stack

Programming language

Python 3.x

IDE / environment

VS Code, Jupyter Notebook, Google Colab

Framework

Scikit-learn (Pipeline, GridSearchCV)

Libraries

NumPy, Pandas, Matplotlib, Seaborn

Database / data

CSV dataset (MySQL optional)

Hardware & OS

Intel Core i5+, 8 GB RAM, 10 GB storage; Windows, Linux, macOS

System Modules / Components

Data Collection

Load CSV with Pandas

Data Preprocessing

Inspection, scaling, split

EDA & Visualisation

Histograms, heatmap, box plots

ML Module

Multinomial LR in a Pipeline

Tuning

GridSearchCV, 5-fold

Prediction / Decision

Crop and probabilities

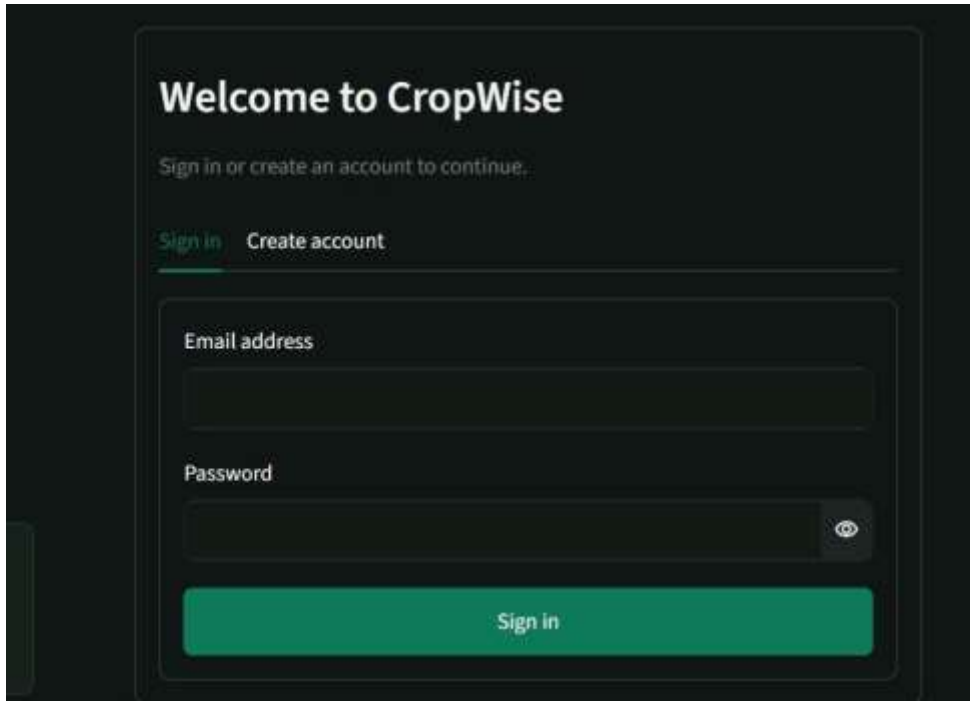
Evaluation

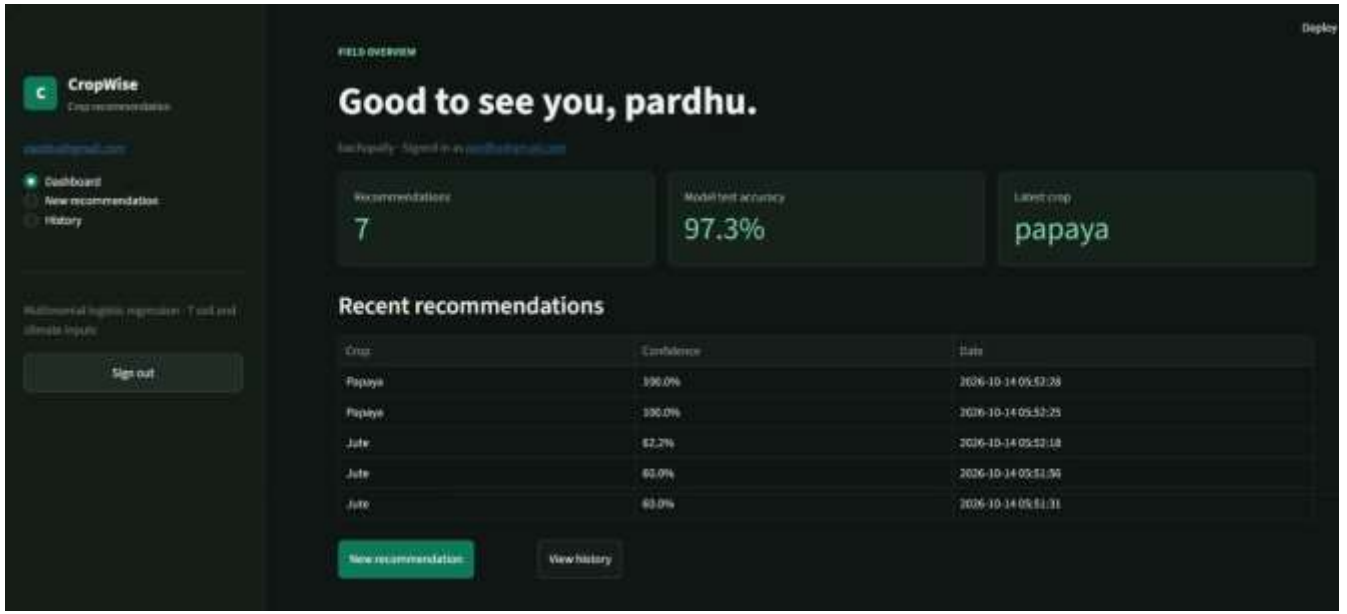
Accuracy, P, R, F1, matrix

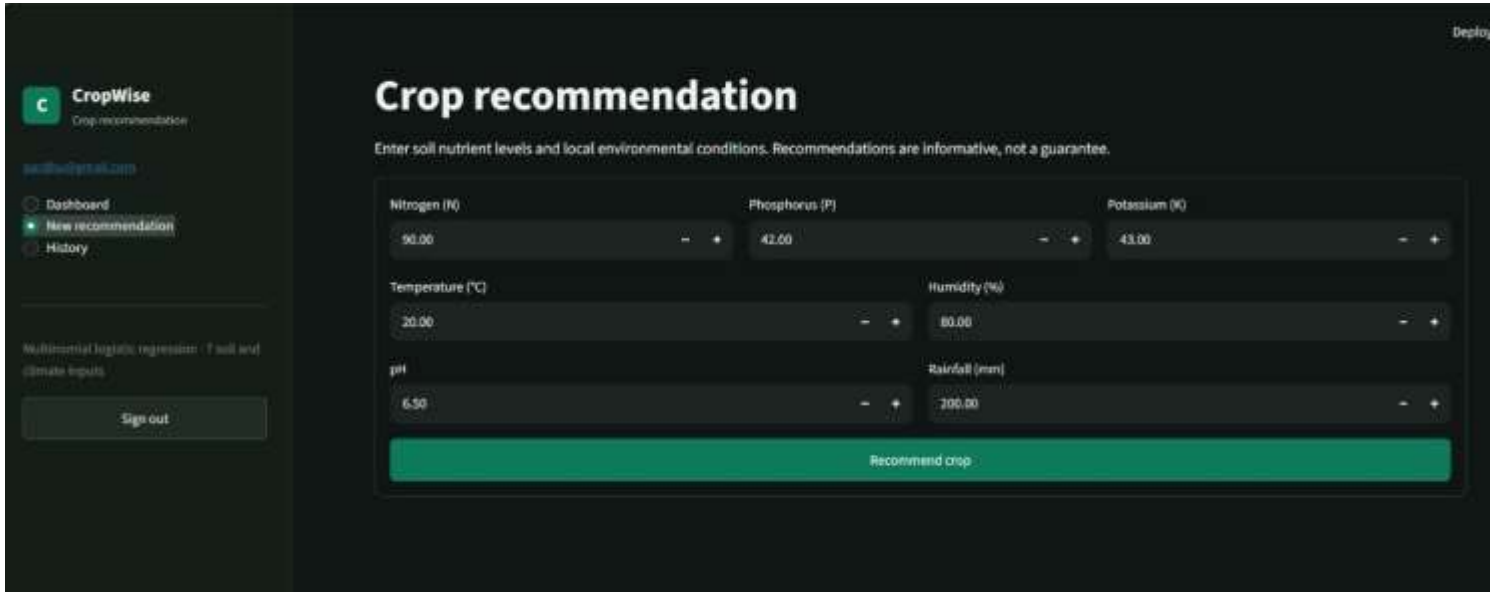
Reporting

Plots and tables

Implementation







Dataset / Input Data

2,200
records

8
columns (7 inputs + label)

22
crops, 100 each

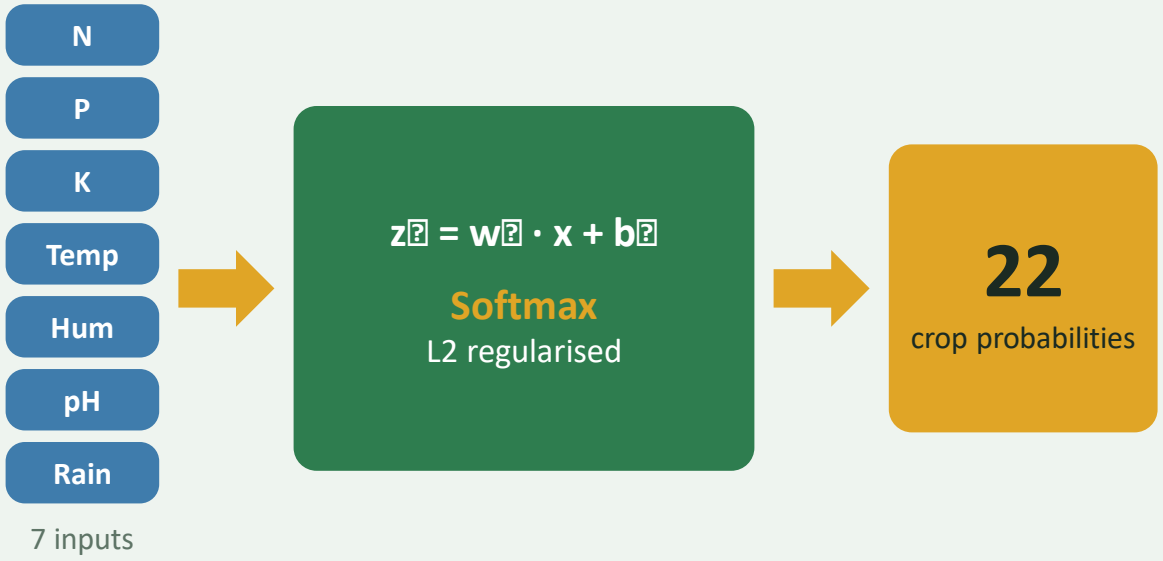
0
missing or duplicate rows

Feature	Range (mean)
N (nitrogen)	0 – 140 (50.6)
P (phosphorus)	5 – 145 (53.4)
K (potassium)	5 – 205 (48.2)
Temperature	8.8 – 43.7 °C (25.6)
Humidity	14.3 – 100 % (71.5)
pH	3.5 – 9.9 (6.5)
Rainfall	20.2 – 298.6 mm (103.5)

Data preprocessing

- No missing values or duplicates to fix
- Outliers kept: genuine crop conditions
- StandardScaler fitted on training data only
- Stratified 80/20 split: 1,760 train, 440 test

Algorithm / Model



Why it was selected

- Handles 22 classes in a single model
- Gives a probability for every crop
- Fast to train and easy to interpret

Important parameters

Solver	lbfgs (multinomial)
Regularisation	L2, C = 100 (tuned)
Max iterations	2000
Scaling	StandardScaler
Train-test split	80/20 stratified (seed 42)
Cross-validation	5-fold stratified
C values tried	0.01, 0.1, 1, 10, 100, 1000

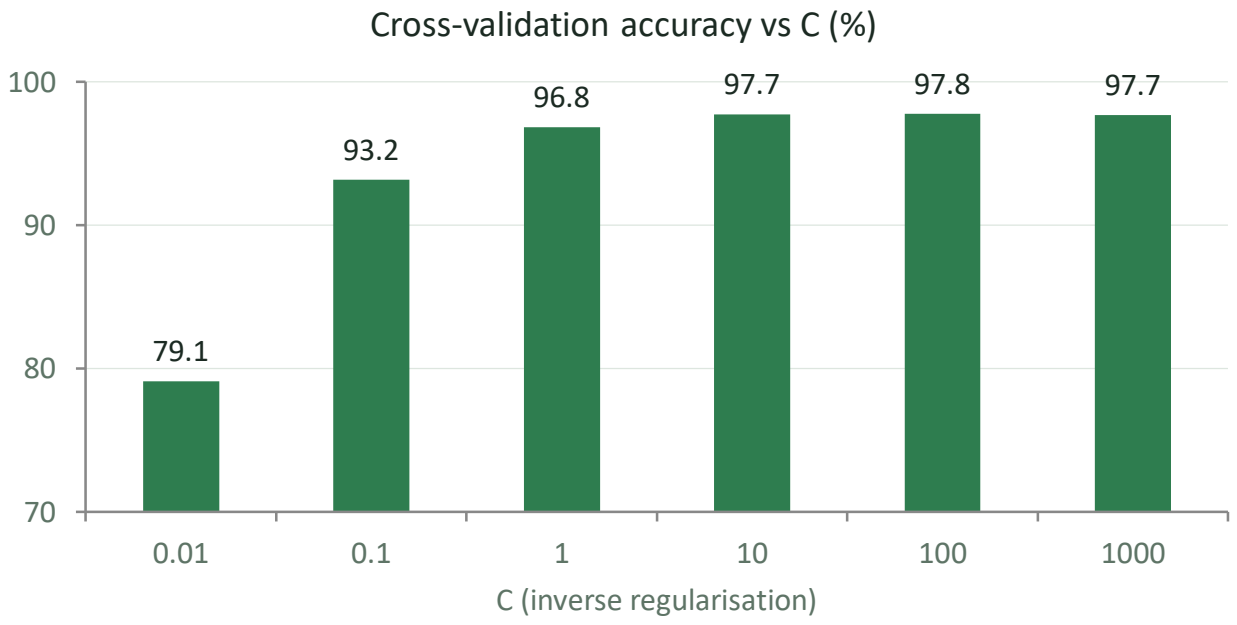
Results: Performance

98.41% Accuracy

98.52% Precision

98.41% Recall

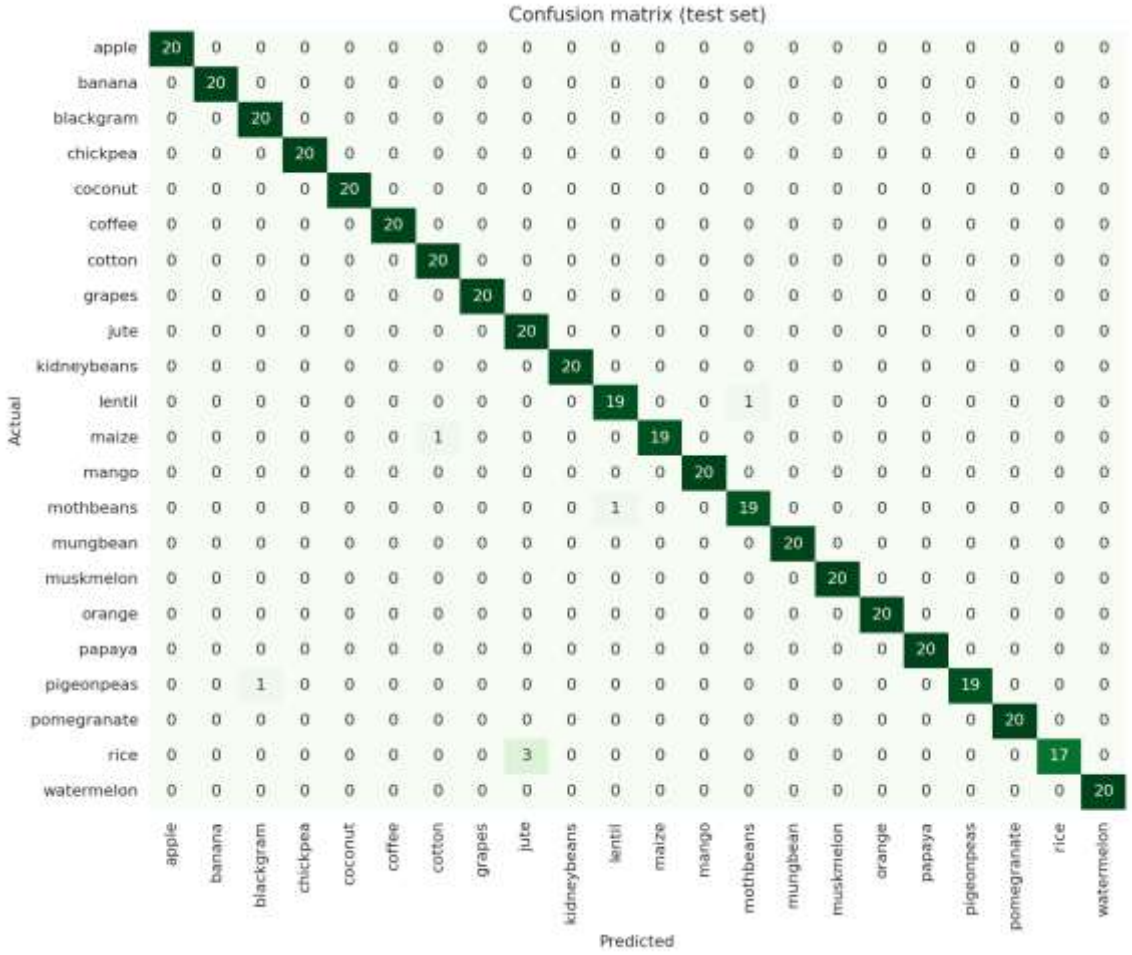
98.40% F1-score



Key findings

- 433 of 440 test samples correct
- 5-fold CV on all data: 98.32% ± 0.18%
- Tuning C lifted test accuracy from 97.27% (C = 1) to 98.41%
- Train 99.03% vs test 98.41%: no serious overfitting

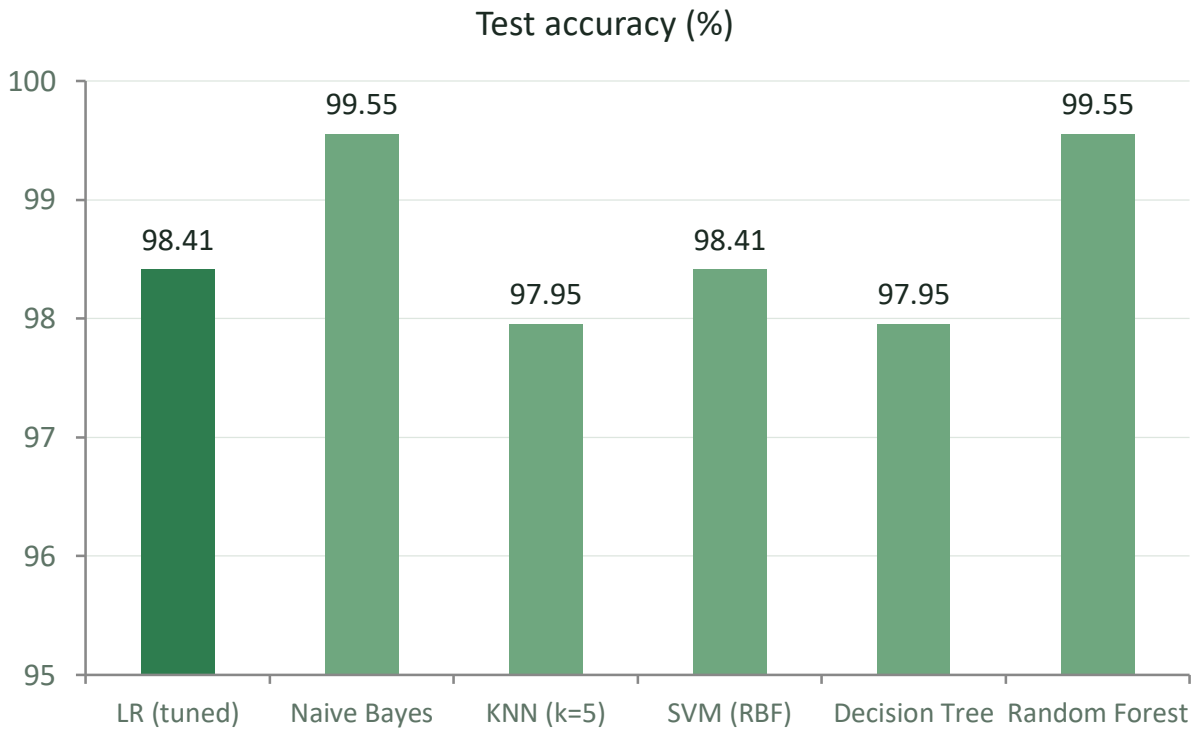
Results: Confusion Matrix



Reading the matrix

- 17 of 22 crops: all 20 test samples correct
- Rice → jute 3 times (both high rainfall and humidity; rice weakest, F1 0.92)
- Lentil ↔ moth beans: 1 each way
- Pigeon peas → black gram: 1
- Maize → cotton: 1

Results: Comparison with Other Methods



Axis starts at 95% to make small differences visible.

Weighted precision, recall and F1 on the same 440 test samples.

Model	Acc.	Prec.	Recall	F1
LR (tuned)	**98.41**	**98.52**	**98.41**	**98.40**
Naive Bayes	99.55	99.59	99.55	99.54
KNN (k=5)	97.95	98.04	97.95	97.93
SVM (RBF)	98.41	98.56	98.41	98.40
Decision Tree	97.95	98.06	97.95	97.94
Random Forest	99.55	99.57	99.55	99.55

Tuned LR ties SVM and beats KNN and Decision Tree. Naive Bayes and Random Forest score higher (99.55%), but LR is fast, interpretable and gives clear class probabilities.

Demo: Input to Output

1 Load data

```
shape: (2200, 8)
N P K temp hum ph rain
90 42 43 20.9 82.0 6.5 203
85 58 41 21.8 80.3 7.0 227
60 55 44 23.0 82.3 7.8 264
label: rice, rice, rice
```

2 Input

```
N = 90
P = 42
K = 43
temperature = 20.9 °C
humidity = 82.0 %
pH = 6.5
rainfall = 203.0 mm
```

3 Processing

```
StandardScaler
  → zero mean, unit
variance
Linear scores z□ for
22 crops
Softmax →
probabilities
Pick highest
probability
```

4 Output

```
Recommended crop:
  RICE

rice     92.7 %
jute      7.3 %
maize     0.0 %
```

A web interface for farmers is planned as future scope.

Advantages

- **Fast and lightweight**
- **High accuracy: 98.41%**
- **Probability for each crop**
- **Interpretable coefficients**

Aspect	Existing	Proposed
Basis	Experience	**Data and ML**
Factors used	A few	**All 7 together**
Output	A guess	**Crop + probability**
Speed	Slow, manual	**Instant**
Accuracy	Not measured	**98.41% on test**

Applications



Farm-level crop planning

Choose a crop before each season from soil and weather data



Agricultural advisory

Support extension workers and soil-testing labs



Agri-tech platforms

Embed in smart-farming apps and services



Education and research

Teaching example of multi-class classification

TARGET USERS Farmers • Agronomists • Agri-tech companies • Students and researchers

Limitations

Current limitations

- Linear model: cannot capture complex non-linear interactions (rice is confused with jute)
- Only 7 inputs: no soil type, season, location, irrigation or market price
- Advisory only: verify against local conditions

Dataset / technical constraints

- Curated, clean, balanced data (100 per crop); real field data is noisier
- One climate value per record: no seasonal variation
- Naive Bayes and Random Forest scored higher here (99.55% vs 98.41%)

Future Scope

Possible improvements

- Compare Random Forest, SVM, XGBoost, neural networks
- Automated hyperparameter search
- Ensemble of models

Additional features

- Soil type, season and location
- Market price and expected yield
- Fertiliser advice for the chosen crop

Deployment / scaling

- Web or mobile app with an input form
- API with live weather feed
- Regional field datasets

Conclusion

Achieved

A crop recommender using Multinomial Logistic Regression on 2,200 records covering 22 crops.

Key outcome

98.41% test accuracy and 98.32% cross-validated accuracy; only 7 of 440 test errors.

Overall contribution

A fast, interpretable ML baseline that supports data-driven crop decisions.

Thank you | Questions?